Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Friendly Neighbourhood Devs
b. Team leader name: Keshav Lohiya
c. Problem Statement: Sustainability & Social Impact

Build an AI-powered solution that addresses a meaningful environmental or societal challenge. Solutions may focus on improving resource efficiency, increasing accessibility, strengthening resilience, supporting communities, or enabling more informed and sustainable decision-making.

Brief about the idea

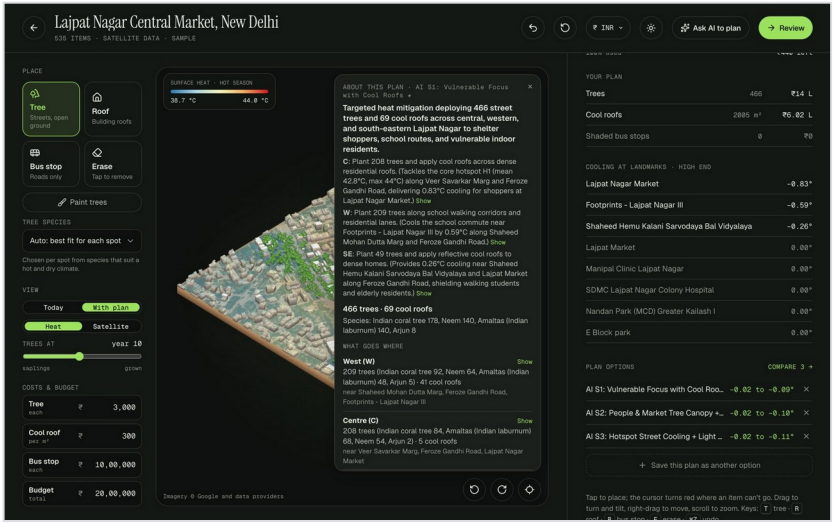
LeafIt is a heat-action planner for city neighbourhoods. Draw an area on a Google satellite map and LeafIt builds a digital twin of it from real satellite and open data. A Gemini agent on Vertex AI tests where trees, cool roofs and shaded bus stops help people most within the budget, then hands back a costed plan, a funding proposal and nearby vendors to quote it.

WHO IT IS FOR

Residents' welfare associations, ward officers, NGOs and CSR teams who spend money on cooling but cannot test a plan before buying it.

WHY THE NUMBERS CAN BE TRUSTED

Every figure comes from a simulator whose coefficients cite published studies. Gemini chooses between options and explains them; every number it writes is checked against the simulator.



REAL RUN, Lajpat Nagar Central Market, New Delhi (0.49 km²)

Gemini's chosen plan for ₹20 lakh: 466 trees and 69 cool roofs. From the twin: 10,307 residents (WorldPop), 25% existing tree canopy, outdoor surfaces 38.7–44.0 °C (Landsat 8/9, Apr–Jun 2023–25).

Opportunities

How different is it from any of the other existing ideas?

Heat maps and dashboards show where a city is hot. LeafIt answers what to do about it, block by block: which mix of trees, cool roofs and shelters, where, for whom, and at what cost per degree.

It starts from the shade already there (1 m canopy height, plus a 150 m ring around the area), so new trees are not wasted where shade already exists.

How will it be able to solve the problem?

It turns guesswork into a tested, costed plan in one flow:

- Brief Gemini in plain words, with a budget
- Gemini runs 2–4 strategies on the simulator and checks each one
- Code refuses a plan with more than 10% of its trees in existing shade
- The plan leaves with a bill of materials by species, a funding proposal and nurseries and contractors within 25 km

USP of the proposed solution

Honest numbers. Cooling is shown as a low–high range from two published studies, every coefficient is cited in the app, and each figure Gemini writes is matched against the simulator.

AI decides, code calculates. Gemini weighs what no single score can: street cooling, cooler homes, shade for people waiting.

Map to money. One session ends in a funding proposal and real vendor quotes.

List of features offered by the solution

01
Digital twin from real data
Landsat 8/9 heat, Open Buildings heights, Meta/WRI 1 m canopy, WorldPop residents by age, OSM roads, Google Places and Air Quality.

02
3D design studio
Place trees, cool roofs and shaded bus stops by hand. The simulator runs in the browser, so every number updates instantly; trees grow in over years 1 to 20.

03
Gemini planning agent
A plain-language brief and a budget. Through five function tools Gemini tests 2–4 strategies and explains its choice zone by zone, naming roads, landmarks and species.

04
Self-check enforced in code
Budget used, trees wasted in existing shade, zones left out, species mix. A plan that wastes trees in shade cannot be recommended.

05
Honest numbers
Cooling is shown as a low–high range from published studies. A Method page cites every coefficient; numbers Gemini invents are flagged to the user.

06
Equity built in
Cooling where children under 5 and adults over 65 live, residents under cool roofs, and shaded bus stops: who benefits, not only the average.

07
Compare and review
Every option Gemini tested lands in a side-by-side comparison. Review sets measured heat today against the plan with a before/after slider.

08
Fund and build
A bill of materials by species, a Gemini-drafted funding proposal, and real nurseries and contractors within 25 km with a ready quote message. 14 currencies.

Process flow diagram or Use-case diagram

USER JOURNEY

1 · Find
Search a place on a Google satellite map

→

2 · Mark
Drag a box, 200 m to 1 km^2

→

3 · Twin
Earth Engine, Maps and OSM data on 10 m cells

→

4 · Plan
By hand, or brief the Gemini agent

→

5 · Review
Compare options, before/after heat, save

→

6 · Act
Materials, funding proposal, vendor quotes

INSIDE STEP 4: THE GEMINI AGENT LOOP (FUNCTION CALLING ON VERTEX AI)

Read the place
get_place_brief, get_zone, plus generated heat and canopy maps

→

Strategise
run_scenario 2–4 times: goal, zones, landmarks, species, roofs, shelters

→

Self-critique
check_plan: budget, trees in shade, zones left out, species mix

→

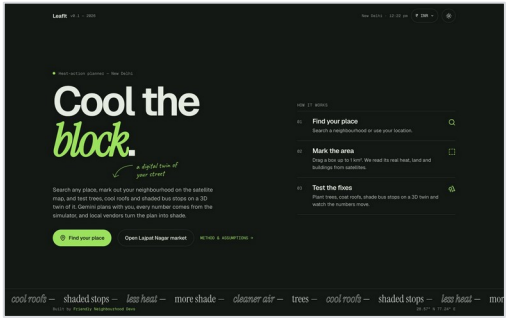
Revise
Choice refused if >10% of trees sit in existing shade (up to 2 rounds)

→

Choose and explain
choose_scenario, then a structured answer zone by zone

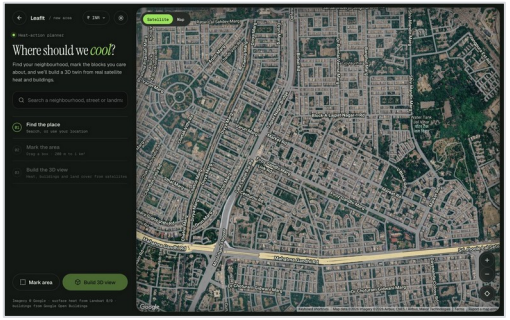
Guardrail: every number in Gemini's answer is matched to the tool outputs; unmatched figures are shown to the user. **Real run (Lajpat Nagar, ₹20 lakh):** 3 strategies simulated, 2 checked, S1 chosen, 0 unmatched numbers, 114 s.

Wireframes/Mock diagrams of the proposed solution (optional)



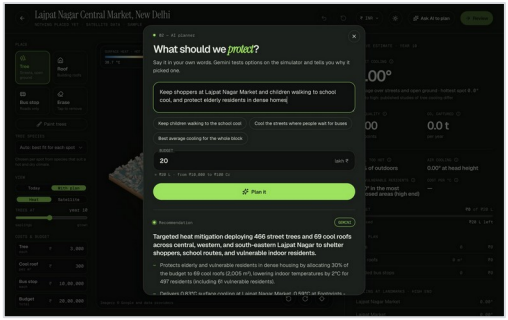
1 · Home

What LeafIt does and where to start; open the sample or find your own place.



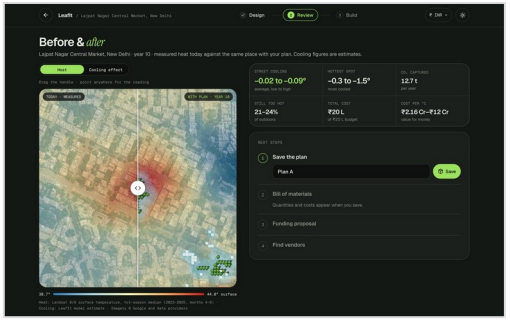
2 · Explore

Search, then drag a box on the Google satellite map to build the 3D twin.



3 · Studio + AI planner

Brief Gemini in plain words with a budget; read its recommendation.

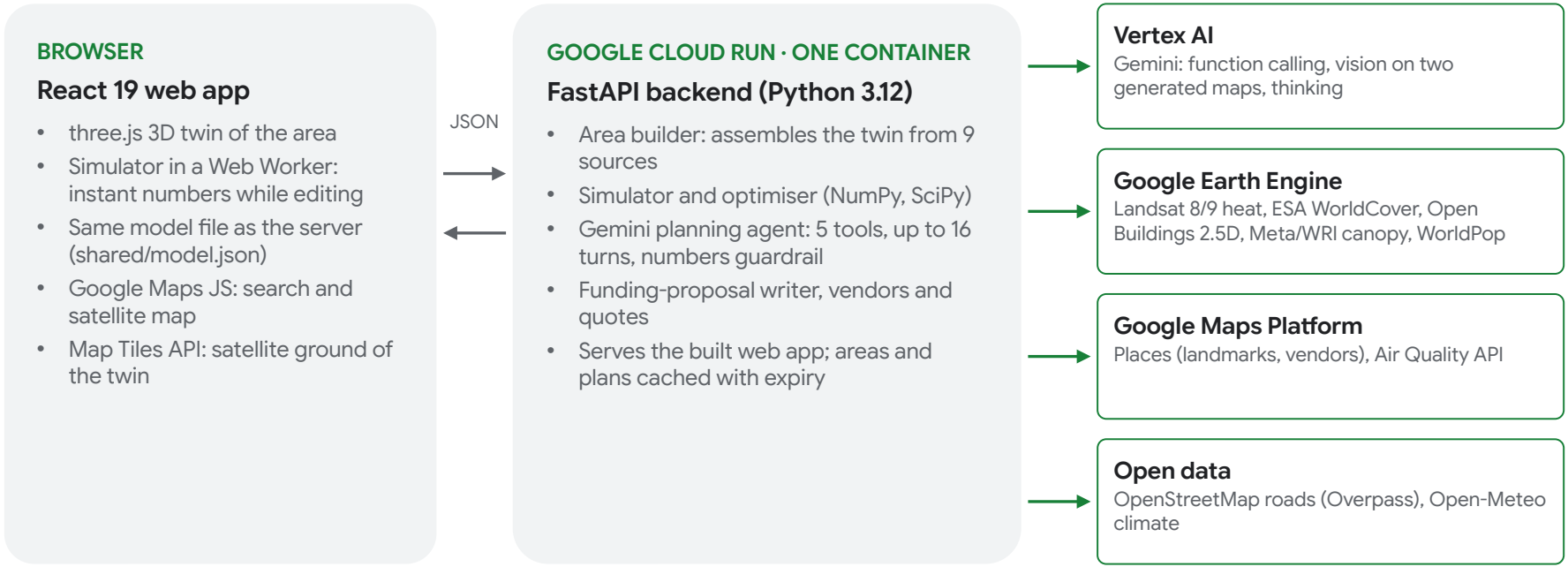


4 · Review

Measured heat today against the plan; cooling as a low–high range.

Screens captured from the working prototype at 1440 × 900, running the version described in this deck.

Architecture diagram of the proposed solution



Delivery: Cloud Build → Artifact Registry → Cloud Run in one script (deploy.sh). Gemini and Earth Engine are called as the service account, so no AI or Earth Engine key ships in the deployment.

Technologies to be used in the solution

LAYER	TECHNOLOGY	USED FOR
Generative AI	Gemini 3.8 Flash on Vertex AI (Gen AI SDK)	Planning agent with function calling, vision and a 1,024-token thinking budget; funding proposals
Geospatial data	Google Earth Engine	Landsat 8/9 surface temperature, ESA WorldCover, Open Buildings 2.5D, Meta/WRI canopy height, WorldPop
Maps	Google Maps Platform	Maps JavaScript API and Places search, Places (landmarks, vendors), Air Quality API, Map Tiles API
Backend	Python 3.12, FastAPI, NumPy, SciPy, httpx, Pillow	Simulator, optimiser, agent tools, generated maps for Gemini
Frontend	React 19, Vite, three.js, Zustand, Framer Motion	3D twin, live simulation in a Web Worker, 14 currencies, light and dark themes
Cloud	Cloud Run, Cloud Build, Artifact Registry, Docker	One container serves the API and the web app
Quality	pytest, Vitest, Ruff, oxlint, shared golden fixtures	157 automated tests; browser and server simulators kept identical

Estimated implementation cost (optional)

WHAT A PLAN COSTS TO BUILD

Street tree ₹3,000 each
Sapling, metal guard, planting, 3 years' care (IISc 2015; MCD 2010)

Cool roof ₹300 per m^2
Cool paint or tiles, Telangana's Cool Roof Policy (2023)

Shaded bus shelter ₹10 lakh each
DTC solar bus-queue shelter; Delhi 2018 range ₹8–20 lakh

EXAMPLE: GEMINI'S PLAN FOR LAJPAT NAGAR, ₹20 LAKH

- 466 trees × ₹3,000 = ₹13.98 lakh
- 2,005 m^2 of cool roofs × ₹300 = ₹6.02 lakh

Total ₹20.00 lakh. All prices are editable in the app, and vendor quotes replace them.

WHAT LEAFIT COSTS TO RUN

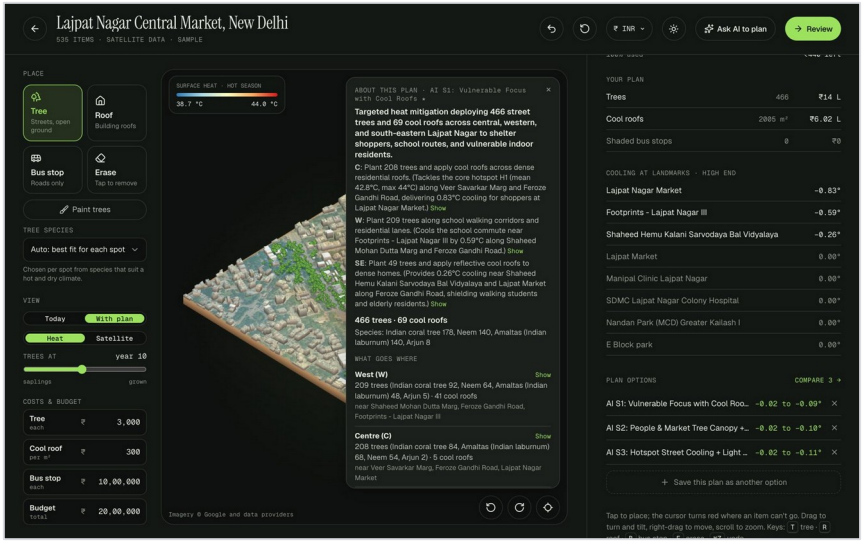
Cloud Run
2 vCPU, 2 GiB, billed per request and scales to zero when idle. No always-on server or database.

Gemini on Vertex AI
Pay per call. A built-in guard caps calls at 600 a day (configurable), so spend cannot run away.

Earth Engine and Maps
Each area is built once and cached, so revisits cost no new satellite or Maps requests.

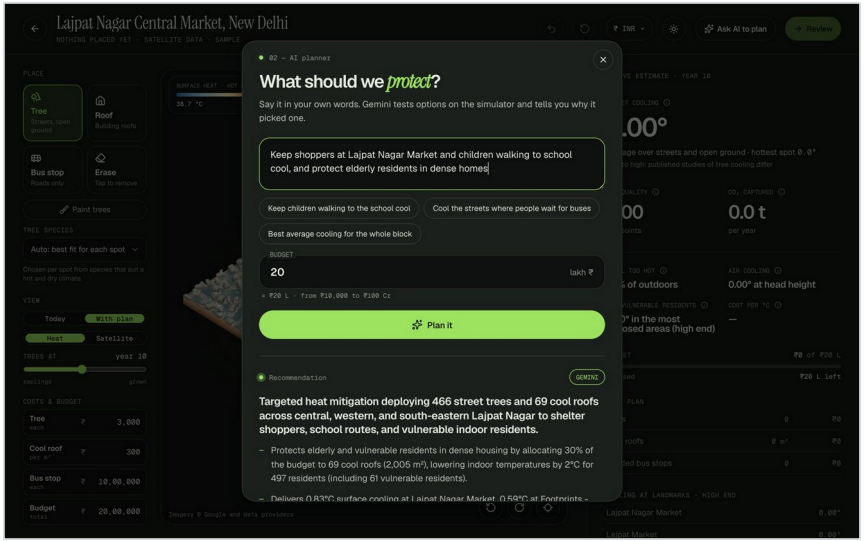
Built on Google Cloud
Usage-based pricing: costs grow only with the number of areas built and plans made.

Snapshots of the prototype



Studio with Gemini's plan applied

466 trees and 69 cool roofs on Lajpat Nagar. "About this plan" names zones, roads and species; landmark cooling and the three options Gemini tested sit on the right.



AI planner

The brief: "Keep shoppers at Lajpat Nagar Market and children walking to school cool, and protect elderly residents in dense homes", ₹20 lakh. Gemini's recommendation follows.

Prototype Performance report/Benchmarking

MEASURED ON THE PROTOTYPE

4.6 ms	Simulator: one plan on a 70 × 70 grid (4,900 cells of 10 m)
0.17–0.19 s	Optimiser: one ₹20 lakh plan
114 s	Gemini planner, brief to answer: 3 strategies, 2 checks, thinking on
0	Numbers in Gemini's answer not traceable to the simulator
157	Automated tests passing (115 backend, 42 frontend), all offline
11 / 11	Shared cases where browser and server simulators agree

GEMINI'S THREE OPTIONS, REAL RUN · LAJPAT NAGAR · ₹20 LAKH · YEAR 10

Option	Street cooling, °C	Hottest spot, °C	Residents under cool roofs	CO_2 / yr
S1 · Vulnerable focus 466 trees · 69 roofs · chosen	−0.02 to −0.09	−0.26 to −1.54	497 (61 vulnerable)	12.7 t
S2 · People and market 533 trees · 46 roofs	−0.02 to −0.10	−0.28 to −1.65	368 (45 vulnerable)	14.5 t
S3 · Hotspots first 600 trees · 30 roofs	−0.02 to −0.11	−0.30 to −1.70	230	16.3 t

Gemini chose S1: the brief asked to protect elderly residents in dense homes, and S1 puts the most residents under cool roofs. Cooling is low to high end of the published range. **Calibration check:** half canopy gives 0.256 °C at the low end (study: 0.268 °C); full canopy cools air 1.00 °C (review: ~1 °C).

Additional Details/Future Development (if any)

01 Narrow the cooling range

Calibrate against Indian measurements: ward air-temperature sensors and afternoon thermal imagery, to replace today's range taken from European and US studies.

02 Indian tree data

Crown size and growth per species from forest departments; size classes are estimates today.

03 Track what gets planted

Record trees and roofs actually installed and their survival, then compare predicted with measured heat each summer.

04 Scale the platform

Move the area and plan cache to Firestore and Cloud Storage. Works wherever Open Buildings has data: Africa, South and Southeast Asia, Latin America.

05 Stronger AI checks

The guardrail confirms every number exists in the tool outputs, not that it is used correctly; add checks on what each figure refers to.

06 Mobile app parity

The Flutter app shares the API; bring it level with the web app, including cooling ranges.

Provide links to your:

GitHub Public Repository

github.com/lohiyakeshav/LeafIt

Demo Video Link (3 Minutes)

To be added

Final Product Link

leafit.cloud.run

Google Cloud H2S

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Build for the world.

Thank you

